FINAL TERM ACTIVITY NO.3 - PYTHON MATPLOTLIB with PANDAS

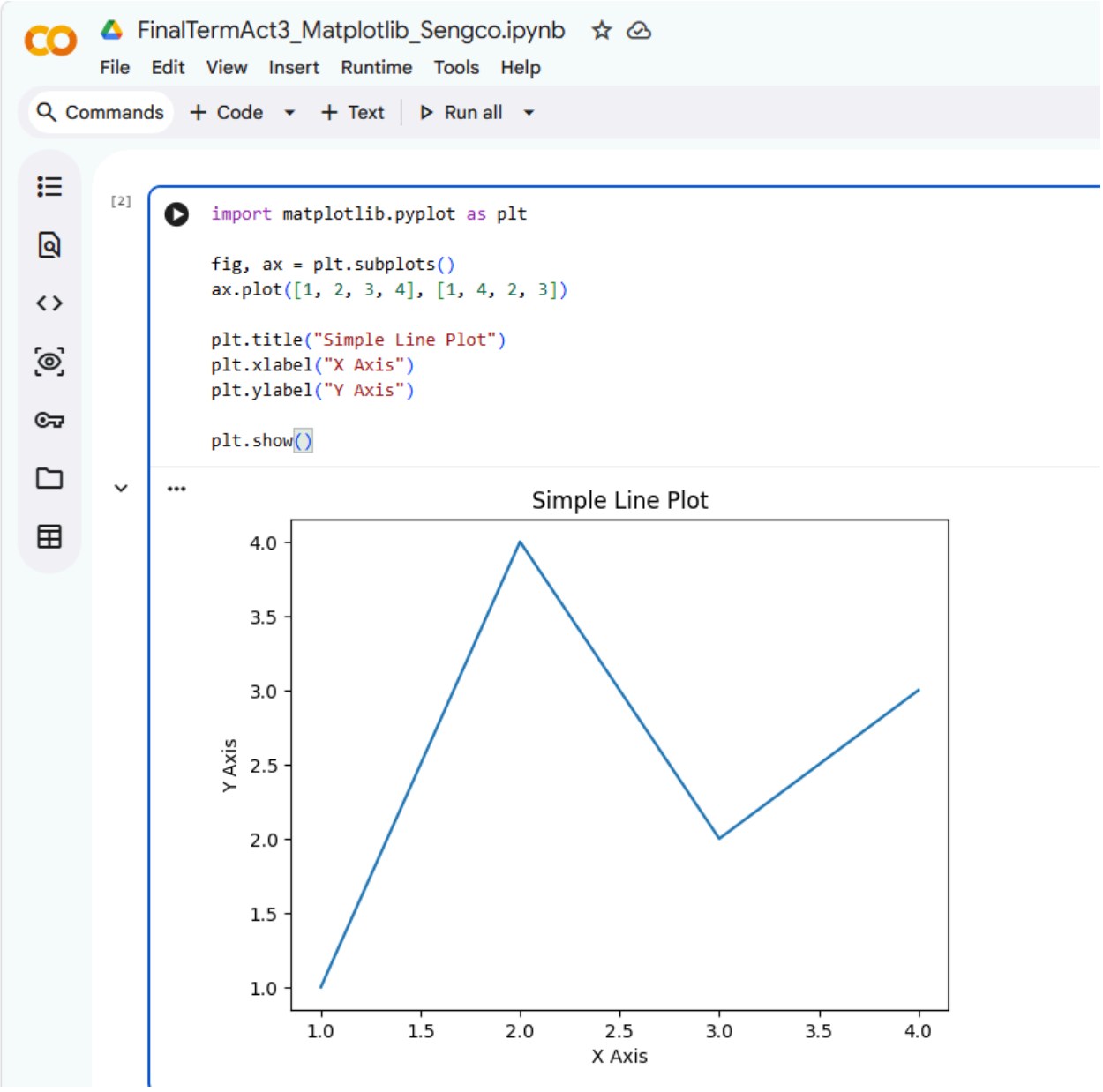


Chart 1: Basic Line Plot

FinalTermAct3_Matplotlib_Sengco.ipynb

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[3]

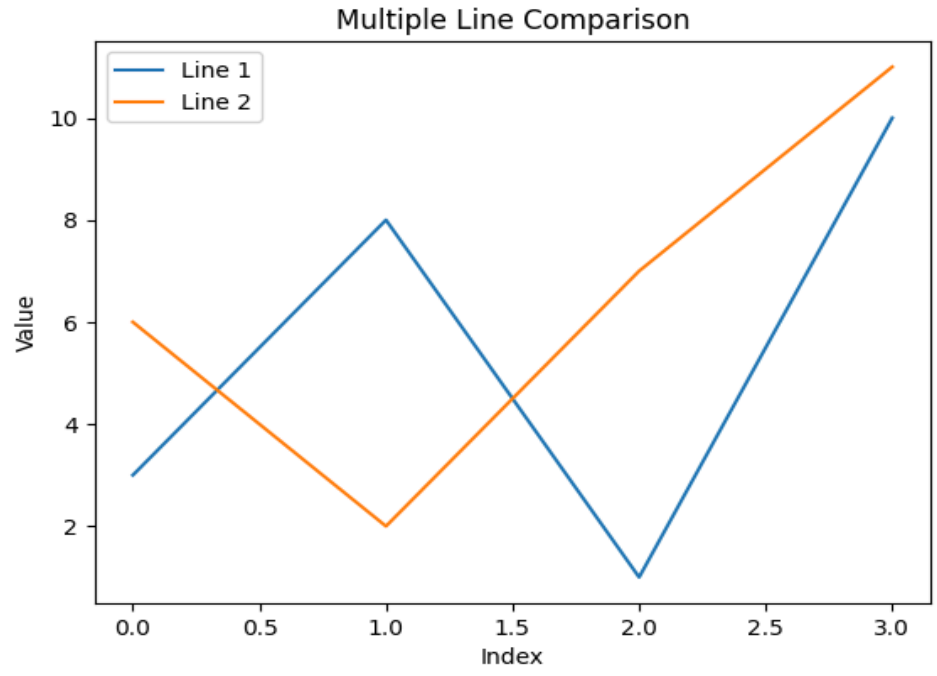
```
import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1, label="Line 1")
plt.plot(y2, label="Line 2")

plt.title("Multiple Line Comparison")
plt.xlabel("Index")
plt.ylabel("Value")
plt.legend()

plt.show()
```



Variables Terminal

Chart 2: Multiple Line Plot

FinalTermAct3_Matplotlib_Sengco.ipynb

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[4]

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5, 7, 8, 7, 2, 17, 2, 9, 4, 11, 12, 9, 6])
y = np.array([99, 86, 87, 88, 111, 86, 103, 87, 94, 78, 77, 85, 86])

plt.scatter(x, y)

plt.title("Scatter Plot of X vs Y")
plt.xlabel("X Variable")
plt.ylabel("Y Variable")

plt.show()
```

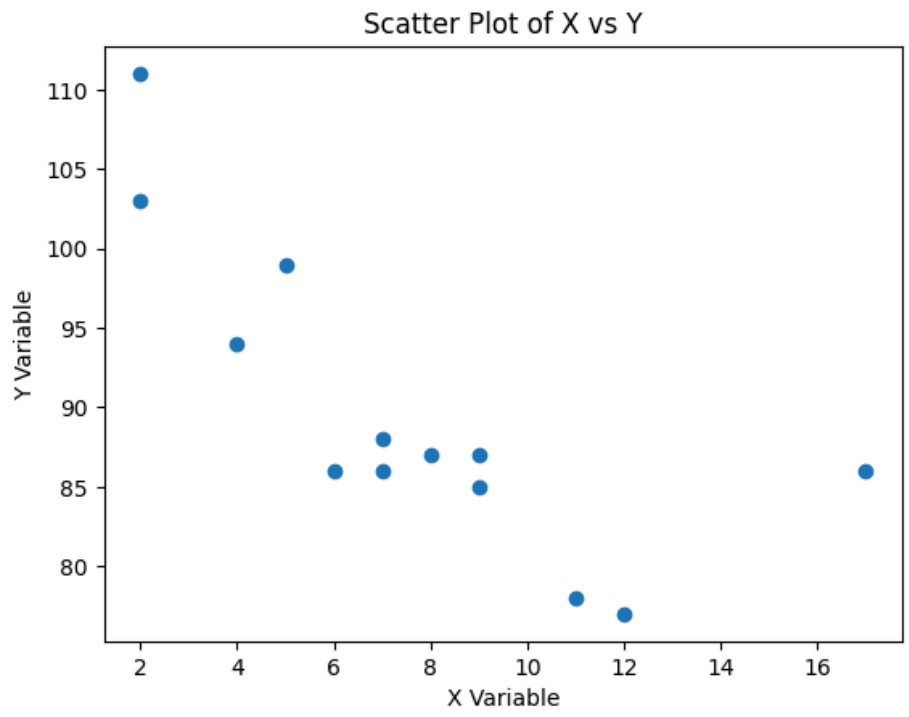


Chart 3: Simple Scatter Plot

FinalTermAct3_Matplotlib_Sengco.ipynb

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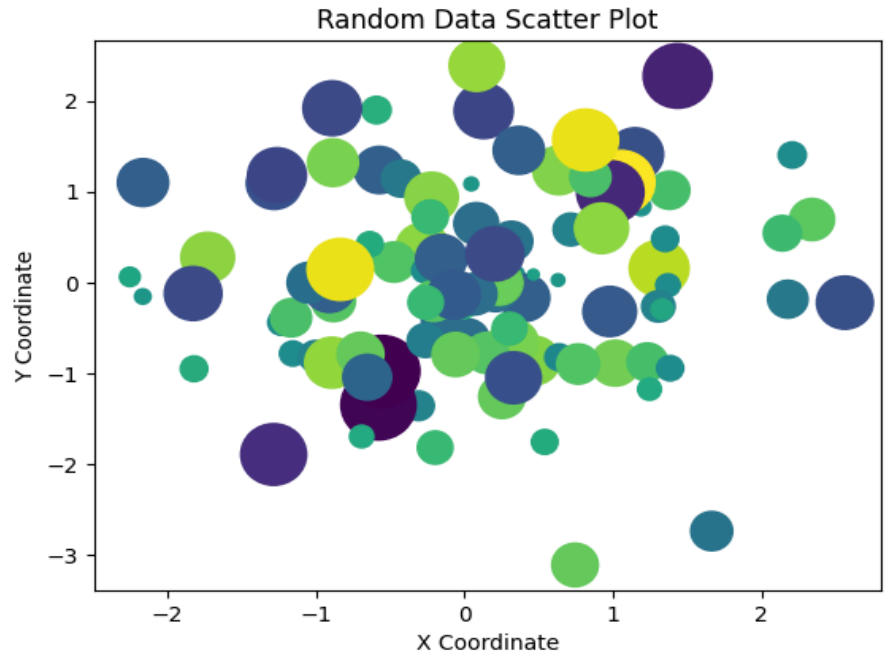
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```
[5]
from matplotlib import pyplot as plt
import numpy as np

x, y, scale = np.random.randn(3, 100)

fig, ax = plt.subplots()
ax.scatter(x=x, y=y, c=scale, s=np.abs(scale) * 500)
ax.set(
    title="Random Data Scatter Plot",
    xlabel="X Coordinate",
    ylabel="Y Coordinate"
)

plt.show()
```



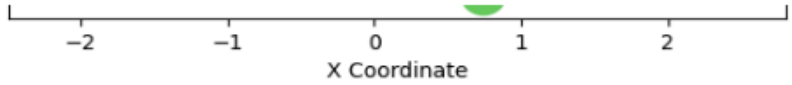
Variables Terminal

Chart 4: Colored Random Data Scatter Plot

FinalTermAct3_Matplotlib_Sengco.ipynb

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[6]

```
import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x, y)

plt.title("Simple Bar Chart")
plt.title("Category Bar Chart")
plt.xlabel("Categories")
plt.ylabel("Values")

plt.show()
```

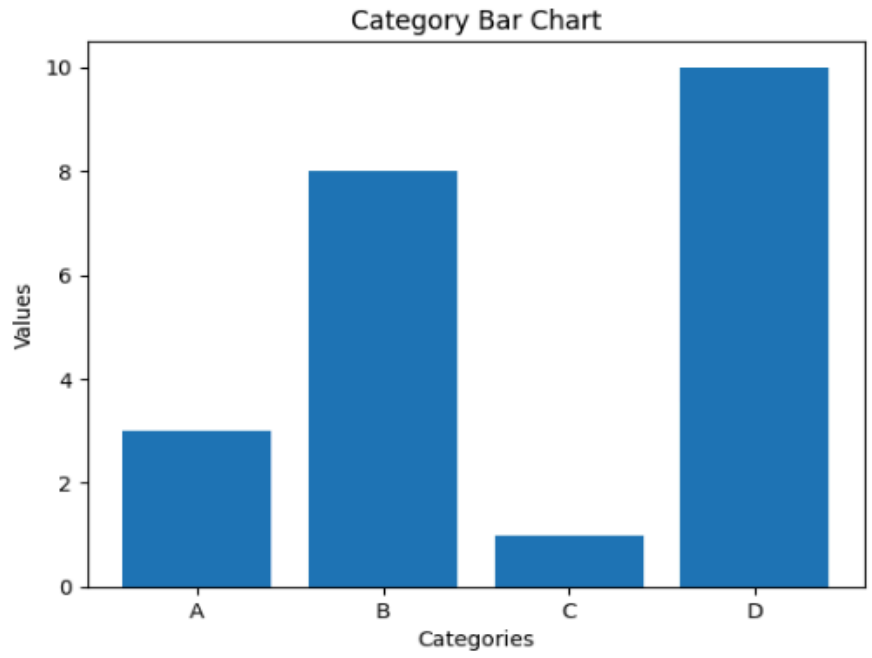


Chart 5: Category Bar Chart

```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.plot(x)


plt.title("Generated Normal Distribution Sequence")
plt.xlabel("Sample Index")
plt.ylabel("Generated Value")

plt.show()
```

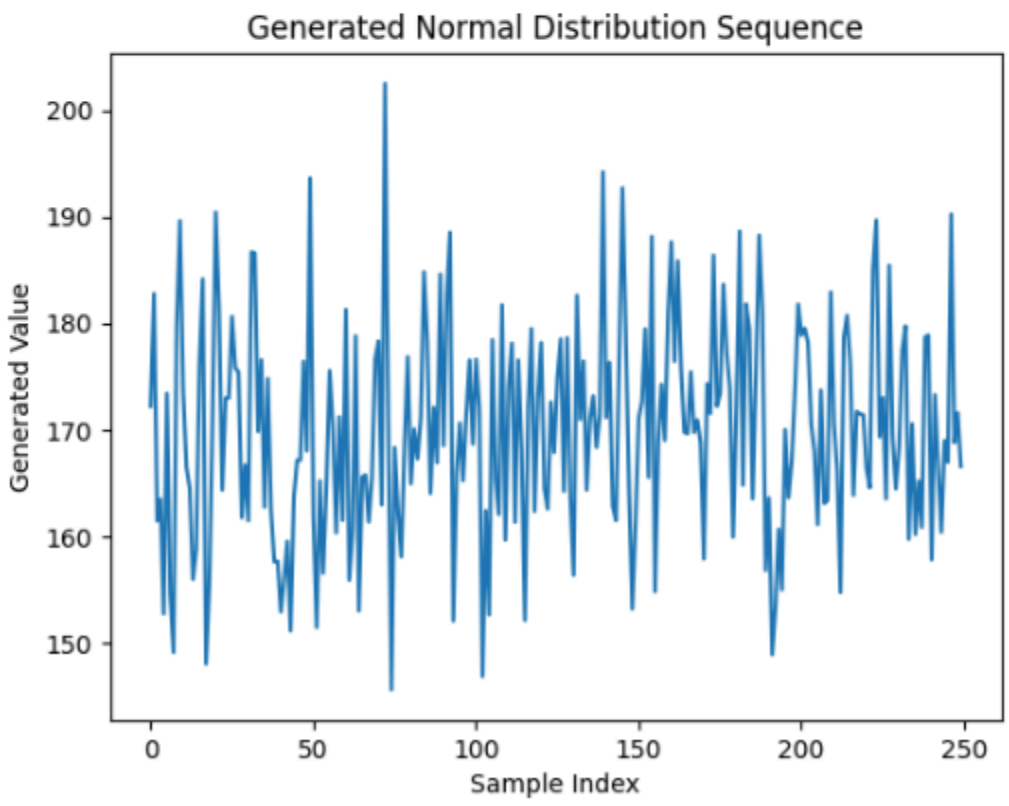


Chart 6: Normal Distribution Values (Line Visual)

FinalTermAct3_Matplotlib_Sengco.ipynb

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Sample Index

[8] 0s

```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)


plt.title("Height Distribution Histogram")
plt.xlabel("Value Range")
plt.ylabel("Frequency")

plt.show()
```

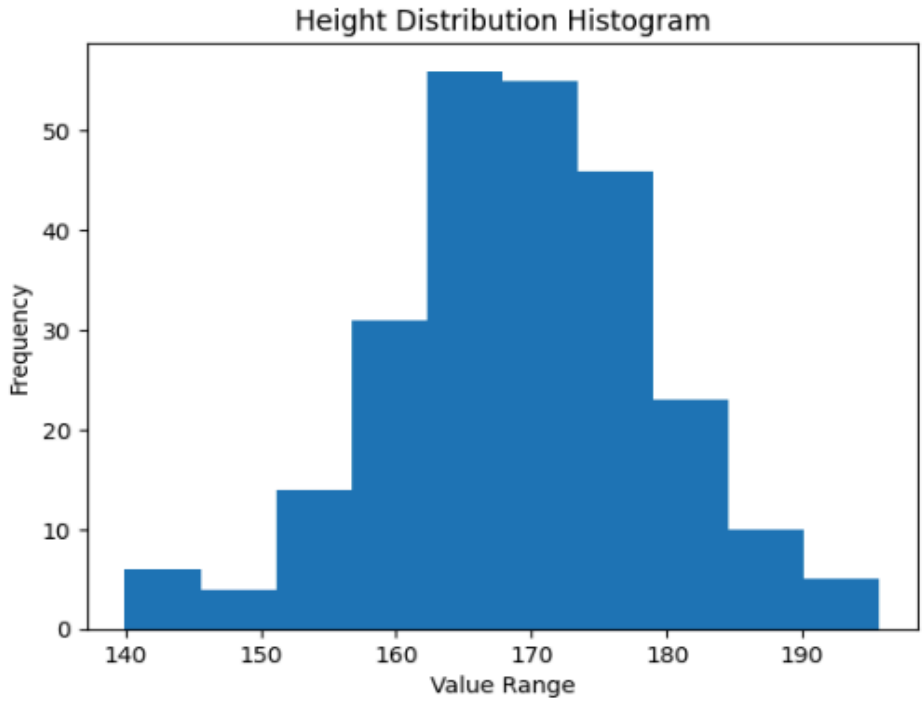


Chart 7: Frequency Histogram

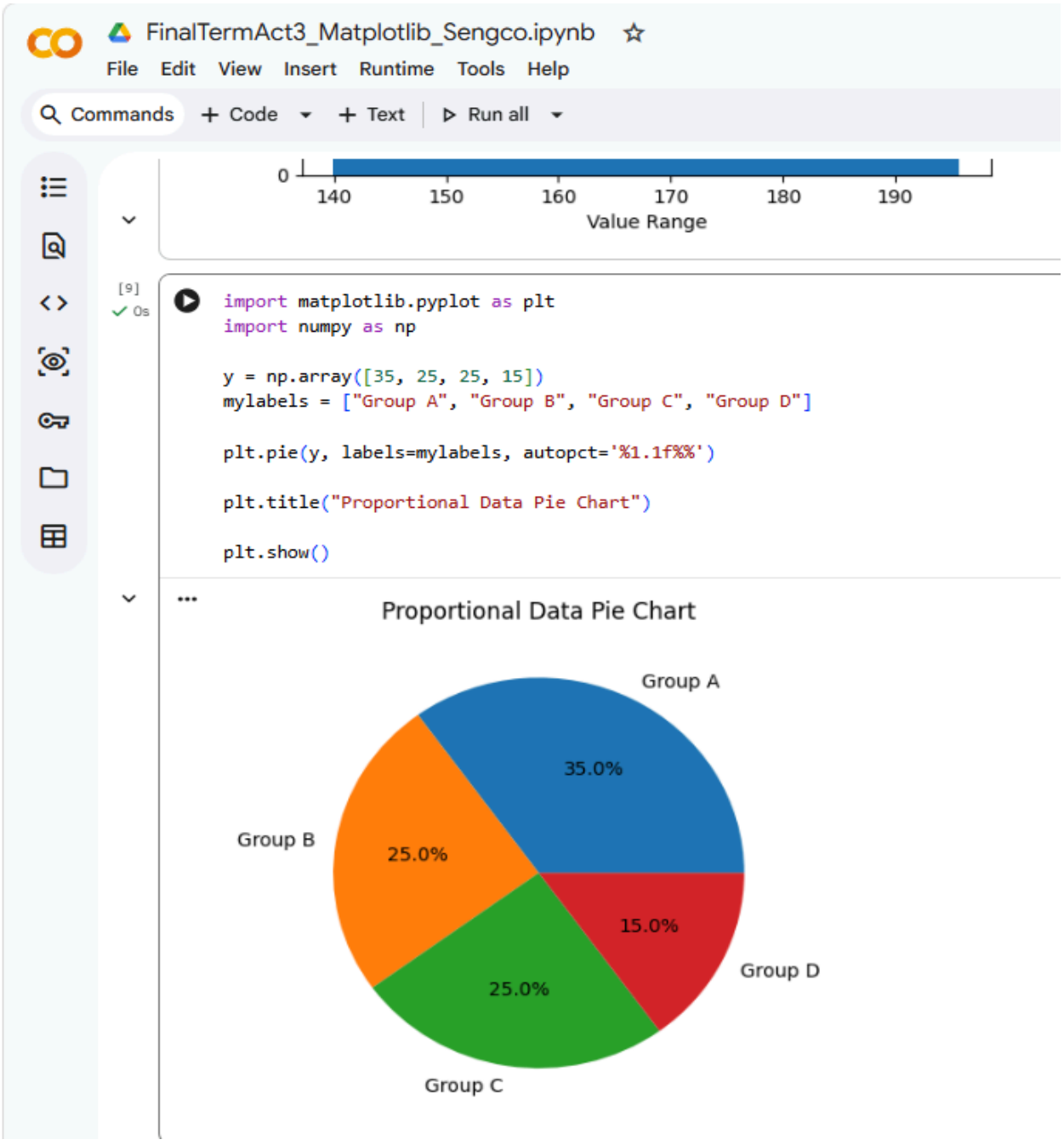


Chart 8: Simple Pie Chart

FinalTermAct3_Matplotlib_Sengco.ipynb Saving...

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[10] 0s

```
import matplotlib.pyplot as plt
import pandas as pd

data = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May'],
    'Sales': [150, 200, 180, 220, 250],
    'Expenses': [90, 110, 100, 130, 140]
}

df = pd.DataFrame(data)

df.plot(x='Month', y=['Sales', 'Expenses'], kind='bar')

plt.title("Monthly Sales vs Expenses (Pandas + Matplotlib)")
plt.xlabel("Month")
plt.ylabel("Amount in USD")

plt.show()
```

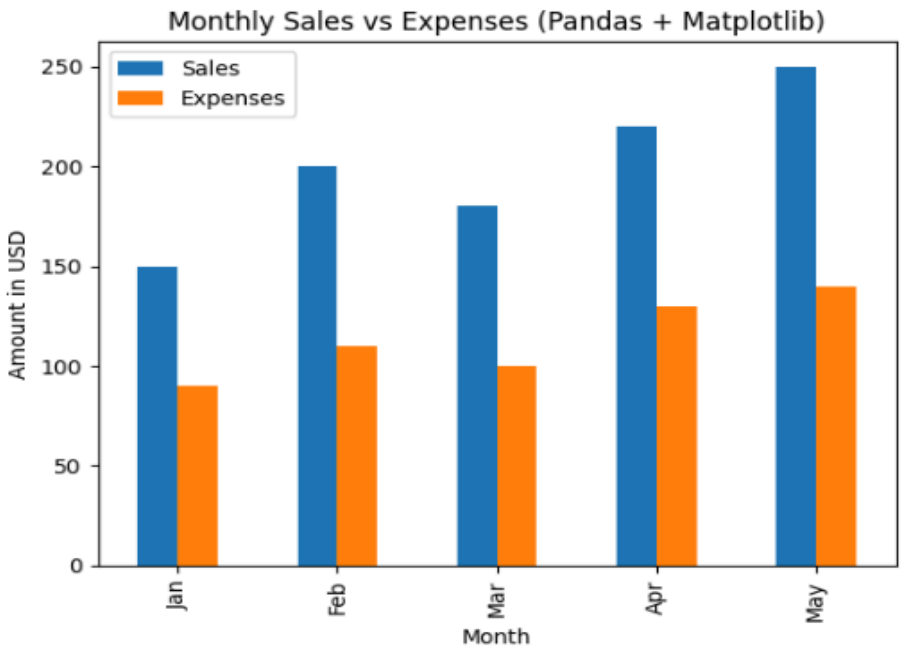
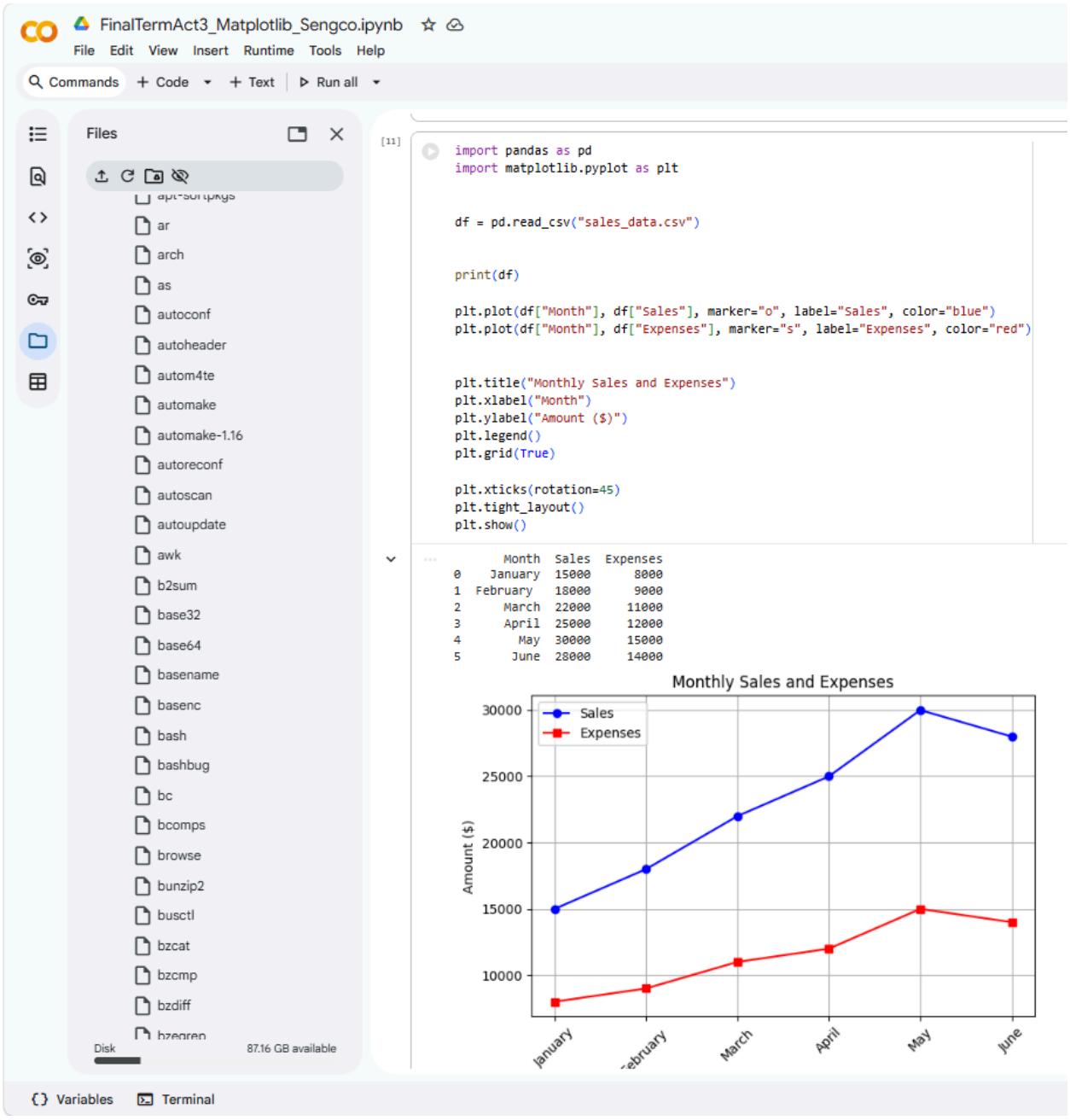
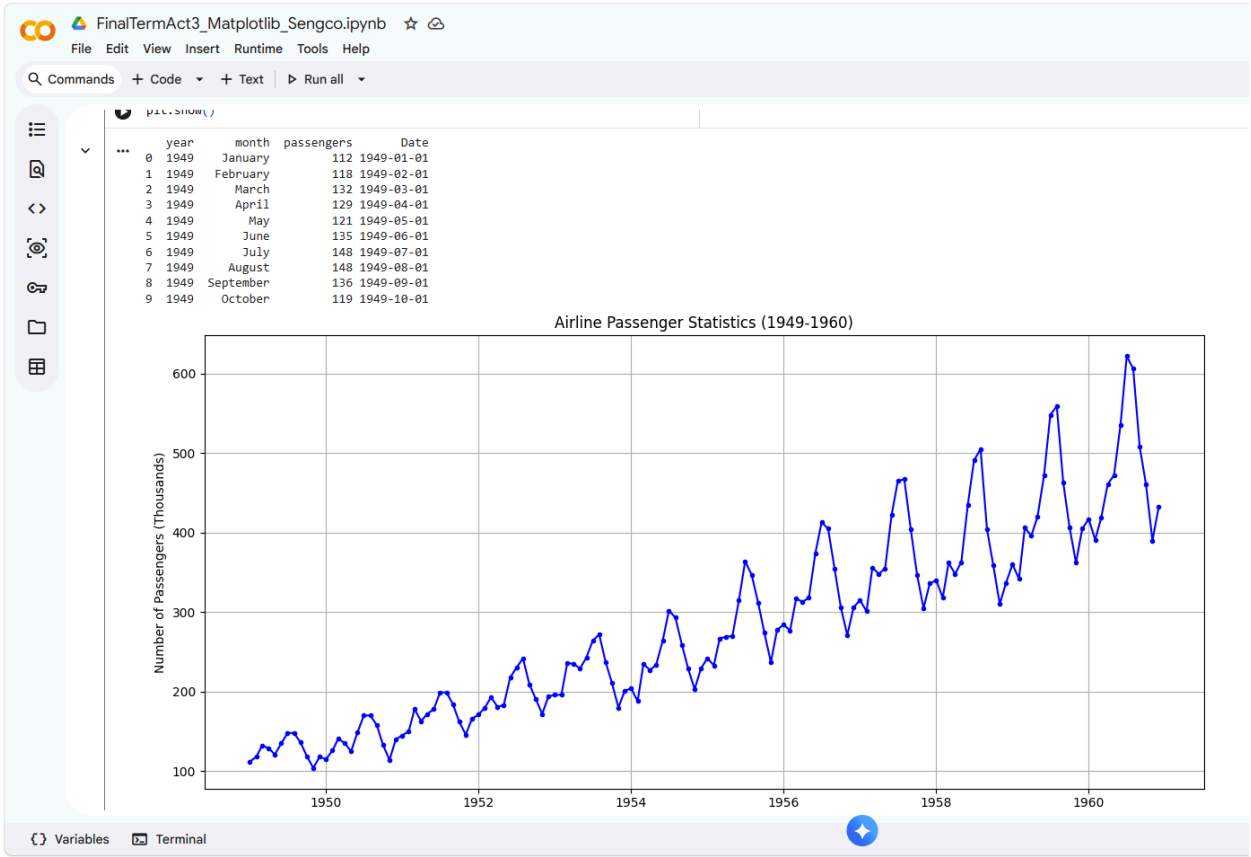


Chart 9: Matplotlib with Pandas DataFrame Plot

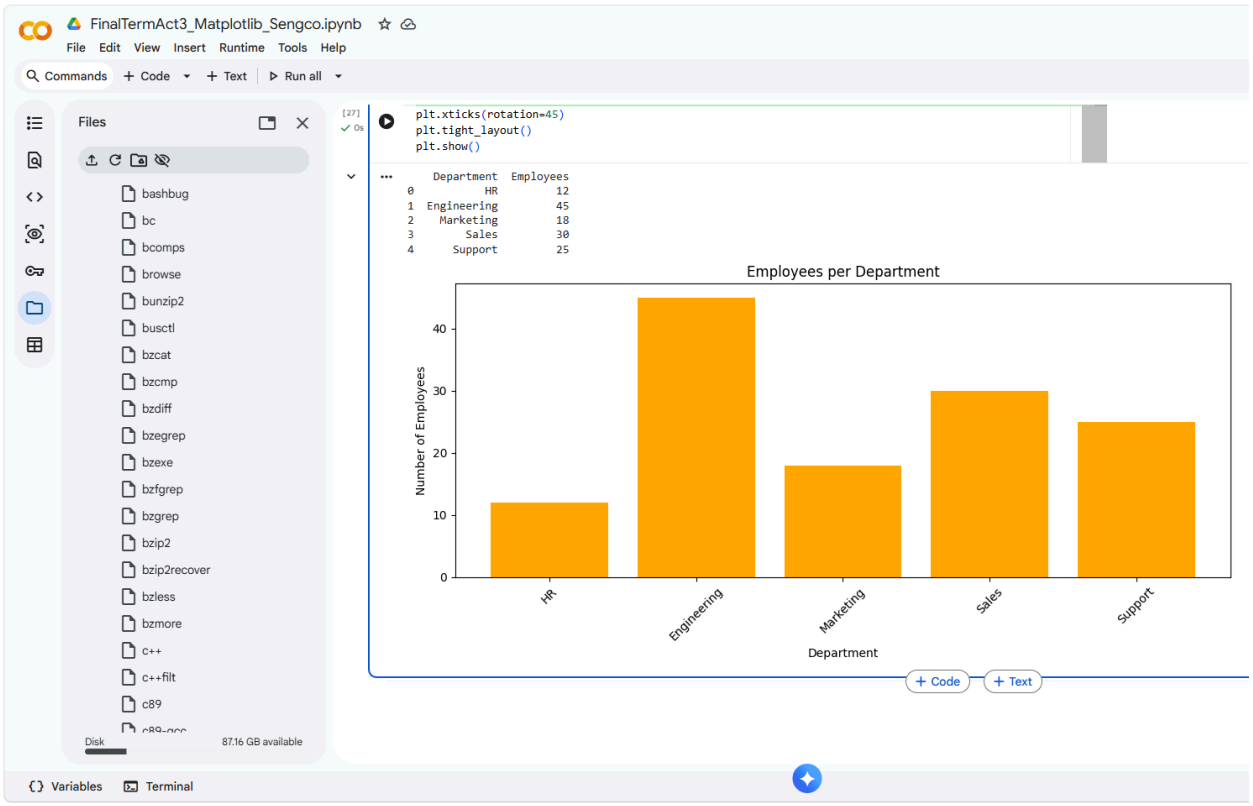
PYTHON Matplotlib with Pandas



Program 1: Create a chart using data from a CSV file



Program 2: Create a chart using data directly from a website



Program 3: Create a chart using data from an Excel file (company_data.xlsx)

Data source	*Python function*	*Description*
CSV file	*pd.read_csv()*	*Reads local CSV files from your environment*
Website CSV	*pd.read_csv(url)*	*Directly loads CSV datasets from a web URL*
Excel file	*pd.read_excel()*	*Reads Microsoft Excel (.xlsx) files*

Program 4: Comparison of the three approaches

PYTHON Matplotlib with Pandas Source Code

```
import pandas as pd
import matplotlib.pyplot as plt


df = pd.read_csv("sales_data.csv")


print(df)

plt.plot(df["Month"], df["Sales"], marker="o", label="Sales",
color="blue")
plt.plot(df["Month"], df["Expenses"], marker="s", label="Expenses",
color="red")


plt.title("Monthly Sales and Expenses")
plt.xlabel("Month")
plt.ylabel("Amount ($)")
plt.legend()
plt.grid(True)

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Program 1: Source Code

```
import pandas as pd
import matplotlib.pyplot as plt


url =
"https://raw.githubusercontent.com/mwaskom/seaborn-data/master/flights.c
sv"

df = pd.read_csv(url)


df["Date"] = pd.to_datetime(
    df["year"].astype(str) + "-" + df["month"],
    format="%Y-%B"
)


print(df.head(10))


plt.figure(figsize=(12, 6))
plt.plot(
    df["Date"],
    df["passengers"],
    color="blue",
    marker="o",
    markersize=3,
    linewidth=1.5
)


plt.title("Airline Passenger Statistics (1949-1960)")
plt.xlabel("Year")
```

```
plt.ylabel("Number of Passengers (Thousands)")

plt.grid(True)


plt.tight_layout()


plt.show()
```

Program 2: Source Code

```
import pandas as pd
import matplotlib.pyplot as plt

# Created dummy data since 'company_data.xlsx' is missing
data = {
    "Department": ["HR", "Engineering", "Marketing", "Sales",
"Support"],
    "Employees": [12, 45, 18, 30, 25]
}
df = pd.DataFrame(data)

print(df.head())

plt.figure(figsize=(10, 5))

plt.bar(df["Department"], df["Employees"], color="orange")

plt.title("Employees per Department")
plt.xlabel("Department")
plt.ylabel("Number of Employees")
```

```
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

Program 3: Source Code